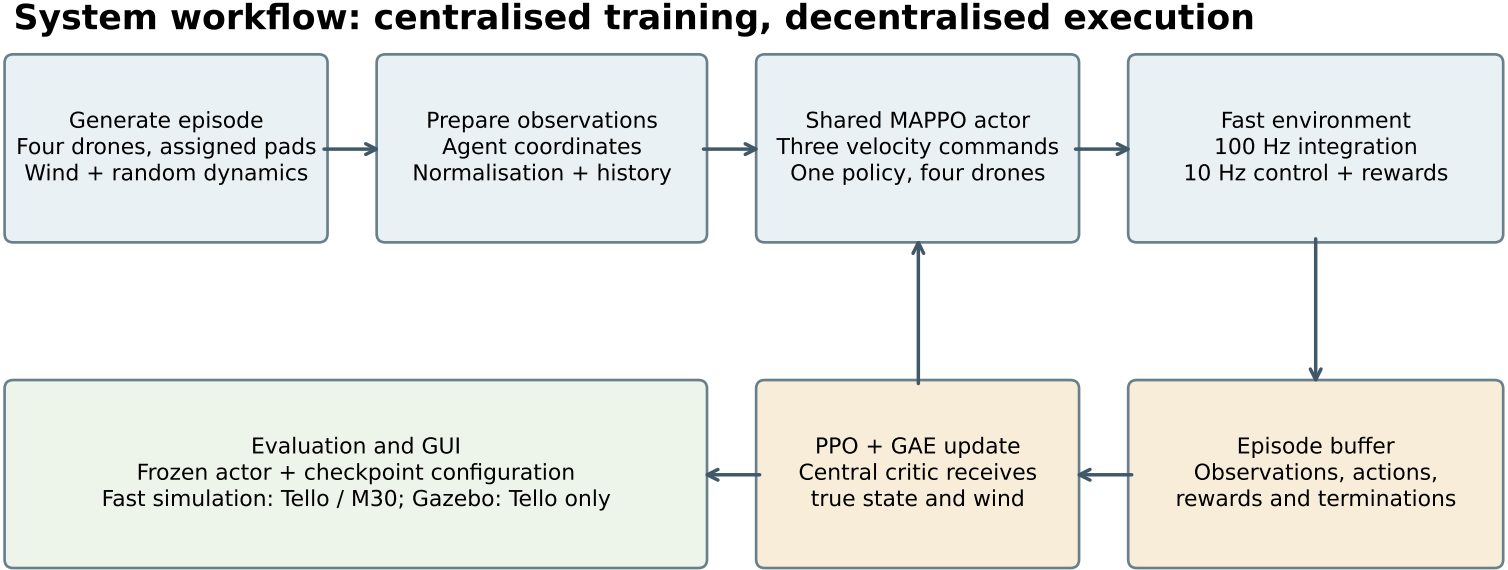
System workflow: centralised training, decentralised execution
Generate episode
Four drones, assigned pads
Wind + random dynamics
Prepare observations
Agent coordinates
Normalisation + history
Shared MAPPO actor
Three velocity commands
One policy, four drones
Fast environment
100 Hz integration
10 Hz control + rewards
Episode buffer
Observations, actions,
rewards and terminations
PPO + GAE update
Central critic receives
true state and wind
Evaluation and GUI
Frozen actor + checkpoint configuration
Fast simulation: Tello / M30; Gazebo: Tello only